Customer Segmentation Analysis

Model Evaluation & Findings Report

Program	CloudExify Data Science Internship — Summer 2026
Module	Month 1 · Project 2 (Final Project)
Domain	Unsupervised Learning — RFM Analysis & K-Means Clustering
Toolset	Python · pandas · NumPy · scikit-learn · matplotlib
Report Date	July 2026

1. Executive Summary

This report documents the customer segmentation analysis performed as the final Project 2 deliverable of the CloudExify Data Science internship. Using RFM (Recency, Frequency, Monetary) analysis combined with K-Means clustering, the transaction history of 199 customers was grouped into three behaviorally distinct segments: High-Value, Mid-Value, and Low-Value.

The resulting segments provide a clear, data-driven basis for differentiated marketing and retention strategy — from VIP treatment for top spenders to targeted win-back campaigns for disengaged customers.

2. Objective & Methodology

The objective was to move beyond simple sales totals and identify distinct customer behavior groups using unsupervised learning, enabling more targeted business decisions than a one-size-fits-all approach. The pipeline followed six stages:

Stage	Description
Data Loading	Loaded raw customer transaction records into a pandas DataFrame; parsed the Date column to datetime.
RFM Calculation	Computed Recency (days since last purchase), Frequency (transaction count), and Monetary (total spend) per customer, relative to a fixed reference date.
Normalization	Standardized RFM features using StandardScaler so no single metric dominates distance calculations.
Optimal K Selection	Applied the elbow method across K = 2 to 10 clusters, plotting inertia to identify the point of diminishing returns.
Clustering	Fit a K-Means model with K = 3 on the standardized RFM data and assigned each customer to a segment.
Profiling & Visualization	Computed per-segment mean RFM values and rendered a 3D scatter plot to visually confirm cluster separation.

3. Dataset Overview

Metric	Value
Total customers analyzed	199
Segmentation method	K-Means clustering (K = 3), on standardized RFM features
Segments identified	High-Value, Mid-Value, Low-Value

4. Segment Distribution

Of the 199 customers analyzed, segment sizes were moderately balanced, with High-Value customers forming the largest single group:

Segment	Customer Count	% of Total
High-Value	88	44.2%
Low-Value	69	34.7%
Mid-Value	42	21.1%

5. Segment Profiles

Average Recency, Frequency, and Monetary values per segment reveal the behavioral fingerprint of each group:

Segment	Avg Recency (days)	Avg Frequency	Avg Monetary (Rs)
High-Value	42.39	9.98	76,564.78
Mid-Value	184.83	5.79	44,631.48
Low-Value	38.42	5.49	40,396.06

Notably, Mid-Value and Low-Value customers have similar spending and purchase frequency, but are separated almost entirely by Recency — Mid-Value customers have not purchased in roughly 185 days on average, compared to just 38 days for Low-Value customers.

6. Key Findings

- K-Means with K = 3 (chosen via the elbow method) produced three clearly separable customer segments based on Recency, Frequency, and Monetary behavior.
- High-Value customers (44.2% of the base) purchase most frequently (~10 transactions on average) and spend the most (Rs 76,565 average), while remaining recently active (~42 days since last purchase).
- Mid-Value customers spend and purchase at levels close to the Low-Value segment, but are distinguished by a much longer Recency (~185 days) — these are customers who were once active but have gone quiet.
- Low-Value customers are the most recently active segment (~38 days) but have the lowest purchase frequency and spend, suggesting newer or lower-engagement customers rather than lapsed ones.
- Recency, not Frequency or Monetary, is the sharpest distinguishing signal between the Mid-Value and Low-Value segments — the two groups spend similarly but differ by roughly 146 days in recency.

7. Business Recommendations

Segment	Recommended Action
High-Value	VIP treatment — exclusive offers, priority service.
Mid-Value	Growth potential — targeted campaigns, loyalty programs.
Low-Value	Win-back campaigns — special discounts, re-engagement offers.

8. Conclusion

The segmentation model successfully partitioned the customer base into three actionable groups using an unsupervised, data-driven approach. The clearest strategic insight is that Recency — not spend or purchase count — is what separates lapsed Mid-Value customers from newer Low-Value customers, meaning retention efforts for these two groups should differ in kind, not just intensity. Combined with the Project 1 sales analysis, this concludes the Month 1 Data Science deliverables for the CloudExify internship.